Graduation Project Proposal

Project: MobiGuide (Market Driven Phone Recommender)

Objective: To analyze smartphone listings from Amazon Egypt, Noon Egypt, and Jumia Egypt; compare prices and specifications; classify phones into four price tiers; and recommend suitable smartphones based on users' budgets and priorities.

Description: The project will build an end-to-end analytics solution for the Egyptian e-commerce smartphone market. Data will be collected from three marketplaces, cleaned and standardized, enriched with engineered features and custom scores, and presented in an interactive Power BI dashboard. The dashboard will include market trends, platform analytics, cross-platform price matching, persona-based recommendations, and an interactive deal finder.

Tools & Design: Power BI dashboard with a consistent visual theme, KPI cards, interactive slicers, price-tier filters, platform comparisons, recommendation tables, and navigation.

Technologies to Use: Python (BeautifulSoup, Selenium, Pandas), Excel/CSV, Power Query (M), Power BI, DAX.

Team Members & Responsibilities:

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- 1-Omar Adel — Scraping: coordinate scraping, build and maintain Python collectors, handle extraction issues, and consolidate source datasets.
- 2-Nourhan Ebrahim — Scraping: support scraping for the three marketplaces, validate extracted fields, manage duplicates, and document collection limitations.
- 3-Marwa Mohamed — ETL & Data Cleaning: clean and standardize fields, parse prices and specifications, handle missing values, and prepare the analysis-ready dataset.
- 4-Mennatollah Wael — Feature Engineering & DAX: implement four price tiers and define Gaming, Camera, Battery, and Value-for-Money scores, plus cheapest-platform logic.
- 5-Kareem Elsayed — Power BI Dashboard & Analytics: build the six dashboard pages, KPI cards, visuals, slicers, navigation, and platform comparison views.
- 6-Mohamed Mahmoud — Insights, Recommendations & Presentation: Validate recommendation outputs, extract business insights, test interactions, and prepare the executive presentation and technical documentation.

Week 1: Project Setup & Data Collection

- Member 1: define scraping approach and build the first extraction workflow.
- Member 2: collect and validate listings from Amazon Egypt, Noon Egypt, and Jumia Egypt.
- Members 3–4: define the required schema, data dictionary, and cleaning rules.

- Members 5–6: sketch dashboard wireframes and agree on key questions and KPIs.

Deliverables:

- Initial raw datasets from the three marketplaces.
- Agreed data schema and data dictionary.
- Initial dashboard wireframe and KPI list.
- Week 1 progress review and issue log.

Week 2: Cleaning, Modeling & Feature Engineering

- Members 1–2: complete collection runs and consolidate the raw data.
- Member 3: clean prices and specification fields, standardize brands/models, and handle missing values.
- Member 4: implement Budget, Midrange, Upper Midrange, and Flagship price tiers.
- Members 3–4: define and implement documented scoring logic for gaming, camera, battery, and value for money.
- Members 5–6: review model fields and confirm that the dataset supports planned visuals and persona filters.

Deliverables:

- Cleaned, consolidated dataset.
- Documented transformation and feature-engineering rules.
- Initial score calculations and data-quality checks.
- Draft Power BI data model.

Week 3: Dashboard Development & Recommendation Logic

- Member 5: build Market Overview, Platform Analytics, and Cross-Platform Price Matcher pages.
- Member 6: build persona pages and the Interactive Deal Finder layout.
- Member 4: implement and test DAX measures, cheapest-platform flags, and recommendation ranking.
- Member 3: verify cleaned fields, relationships, data types, and refresh behavior.
- Members 1–2: investigate scraping gaps and correct source-data issues where feasible.

Deliverables:

- Working draft of the six-page Power BI dashboard.
- Tested DAX measures and recommendation logic.
- Initial market and platform findings.
- Issue list and fixes prioritized for final validation.

Week 4: Validation, Storytelling & Final Submission

- Members 1–2: document scraping methodology, source coverage, and limitations.
- Members 3–4: validate calculations, price tiers, score behavior, and cross-platform matches.
- Members 5–6: refine dashboard design, write insight narratives, and prepare the presentation.
- All members: conduct end-to-end testing using multiple budgets and user priorities,
  prepare the project demo and practice answers to evaluation-board questions.

Deliverables:

- Final interactive Power BI dashboard (.pbix).
- Executive presentation deck (.pptx).

- Technical documentation covering data collection, cleaning, data model, scoring, and limitations.
- Final validation checklist and project demonstration.